

DIP Project Proposal

Edge-Aware Image Fusion through Segmentation and Blending

Team

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Introduction

Have you ever taken two photos and wondered if you could combine the best parts of both — sharp details from one, and rich colors from the other — without messy seams or blurry edges? This project aims to design an edge-preserving image fusion pipeline that segments important regions, blends them seamlessly, and preserves edges for natural-looking results.

By combining the strengths of established segmentation techniques with advanced blending strategies, our approach will produce images that are visually smooth, structurally consistent.

Central Project Idea

We propose a three-stage pipeline: Segmentation → Blending → Edge-preserving Refinement

- **Segmentation (Graph Cuts)**

- Image segmentation will be implemented using graph cuts (2).
- In this method, each pixel is treated as a node in a graph, with edges weighted according to similarity between neighboring pixels.
- By applying a min-cut/max-flow algorithm, the image can be optimally divided into foreground and background regions or other meaningful partitions.
- This provides precise control over which regions will be blended.

- **Blending (Poisson Image Editing)**

- We will apply multi-resolution gradient-domain (Poisson) blending to combine the segmented regions. (1)
- This method matches gradients (intensity changes) between source and target regions.
- From this, we will acquire smooth transitions at boundaries by avoiding sharp seams, mismatched lighting, or color bleeding which are common problems.

- **Edge Preservation (Guided Filtering)**

- We want to integrate edge-preserving filters (bilateral filtering or guided filters - yet to decide based on requirements (3)) to maintain sharp contours and avoid halos.
- Bilateral filtering preserves edges by combining spatial closeness and intensity similarity, but it can be computationally heavy. It produces sharper images.

- Guided filtering uses a guidance image to decide how pixels should be averaged, making it faster and efficient.
 - This step will ensure that important structural details remain intact.
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Ablations

- **Segmentation (Graph Cuts):** We will vary color models (RGB/HSV/Lab) and neighborhood systems (4- vs 8-connected) to see how they affect mask accuracy and boundary sharpness.
 - **Blending:** Although Poisson blending is theoretically stronger for gradient consistency, we will compare it with Laplacian pyramid blending (levels = 3/5/7) and also test seam-band widths (7/11/21 px) to check if theory holds up in practice.
 - **Edge-Preserving Filters:** Guided filtering ($r = 8/16$, $\epsilon = 1e-3/1e-2$) and bilateral filtering ($\sigma_s = 4/8/16$, $\sigma_r = 0.05/0.1/0.2$) will be tested both globally and in narrow seam bands to balance detail preservation and runtime.
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Applications

This project has broad real-world value:

- Photography/Creative Media → Seamlessly combining exposures or textures for HDR imaging, panorama stitching, or artistic photo edits.
 - Medical Imaging → Fusing MRI and CT images to combine structural and functional information while preserving boundaries between tissues.
 - Remote Sensing → Combining multispectral satellite images (e.g., thermal + optical) for clearer, edge-preserving environmental analysis.
 - Augmented Reality → Blending virtual objects into real scenes without jagged edges or visible seams.
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References

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